2023-2024
3rd year /Electronic III
Electronic and Telecommunication Department.
Engineering Science/ Kufa University

Electronic Design Lab course

1. Introducing EAGLE

EAGLE is a software application that enables the design of circuit boards and below are some of their features:

Component library: A collection of devices that can be included in a design. EAGLE will have most of the parts needed. If not, the site http://www.cadsoftusa.com/downloads/libraries provides more libraries for free download. If part Still can't find it. Chapter 8, "Creating Libraries and Components," explains how to custom design Pieces.

Schematic editor: Allows you to draw an initial circuit design.

Board editor: Editor defines the layout and orientation of the actual circuit board. After the schematic design is complete, then a board file (*.brd) that defines the layout of the actual circuit board.

Device editors: Used to design new components. Make it possible to design a new one. This process consists of three steps:

1. Create a design for the schematic editor. This is called a symbol.
2. Create a design for the panel editor. This is called a package.
3. Create a link between the icon and its package. This is called the device.
4. Autorouter: A tool that automatically determines how connecting circuit elements.

5. CAM (Computer Aided Manufacturing) processor: tool that reads the board design and produces files for manufacturing the boards.

2. Circuit boards

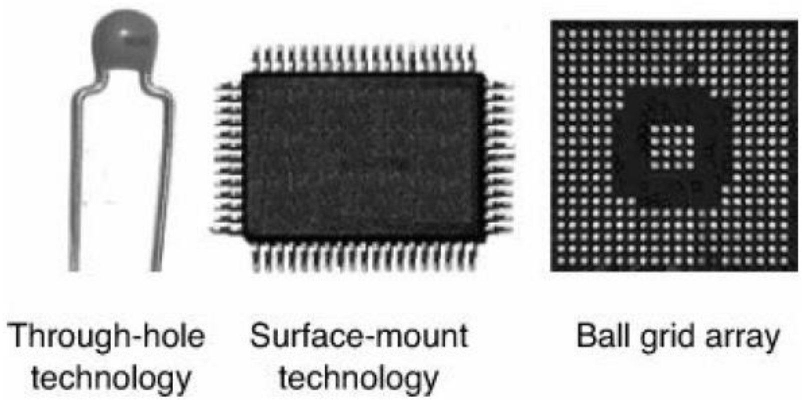
Circuit boards are usually divided into three categories: single-sided, double-sided, or multi-layer. A circuit board can provide mechanical support and electrical connections between components

There are five steps to PCB design.

1. Create a new project.
2. Design a diagram that shows the circuit components and their connections.
3. Create a circuit board design from the schematic and place the corresponding enclosures for the circuit components.
4. Routing communications between packets.
5. Convert the board design into files that can be sent to the manufacturing facility.

The components are classified according to three common types of terminals (leads).

- Through-Hole - Leads are wires that go into holes in the board.
- Surface mount technology (SMT). Connectors are metal tabs mounted on the perimeter of the device.
- Ball Grid Array (BGA) – The leads are metal balls located at the bottom of the device.



The location on the board that comes into contact with the component wire is called the pad. Circuit boards use conducting lines called traces. The process of establishing board traces is called routing.

3. Designing the Circuit Board

The circuit design process in the board editor consists of three steps:

1. Adjust the circuit board size.
2. Place each component on the board.
3. Establish connections between components. Moving components in the panel editor is exactly the same.

Electronic Design Lab 01

Design of inverted amplifier

The LM741 is used as an operational amplifier, the value of R1 should be set to 3.3kΩ and the value of R2 should be set to 1.1kΩ

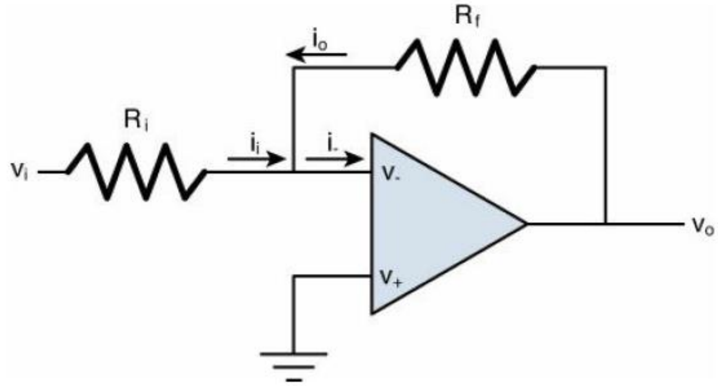


Figure 1: An Inverting Amplifier Circuit

Design process steps:

1. Create the project
2. Design the diagram.
3. Place the components in the board editor.
4. Route the connections between them, and create Gerber and Excellon files.

The basic inverted amplifier design has only five components:

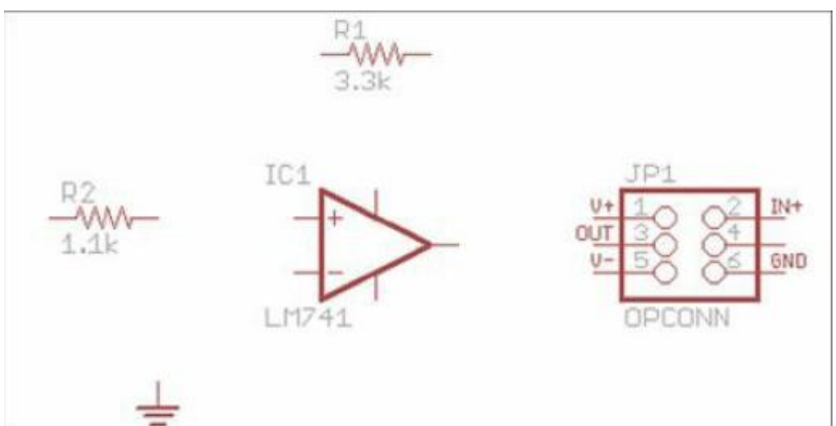


Figure 2: Arrangement of Symbols in the Schematic Editor

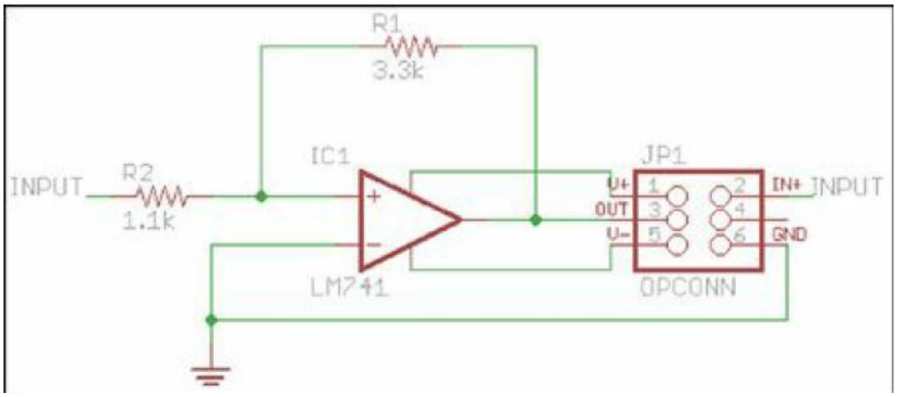


Figure 3: Full Schematic of the Inverting Amplifier Circuit

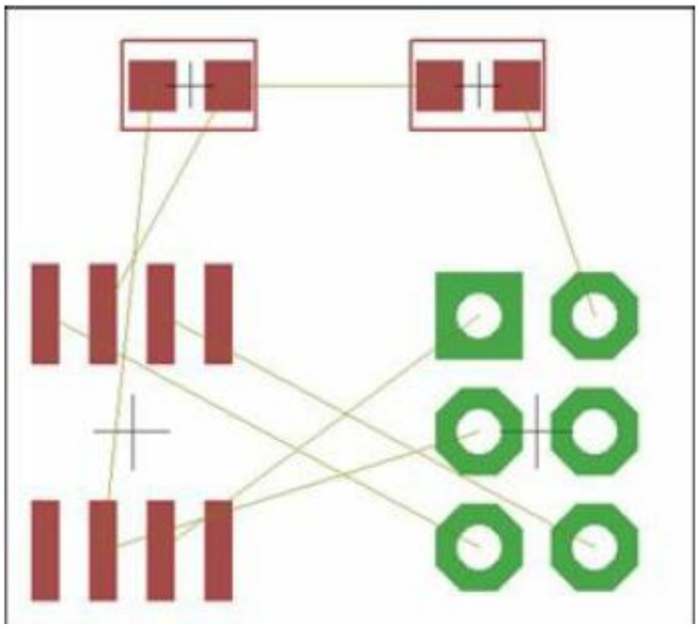


Figure 4: Board Layout – Inverting Amplifier Circuit

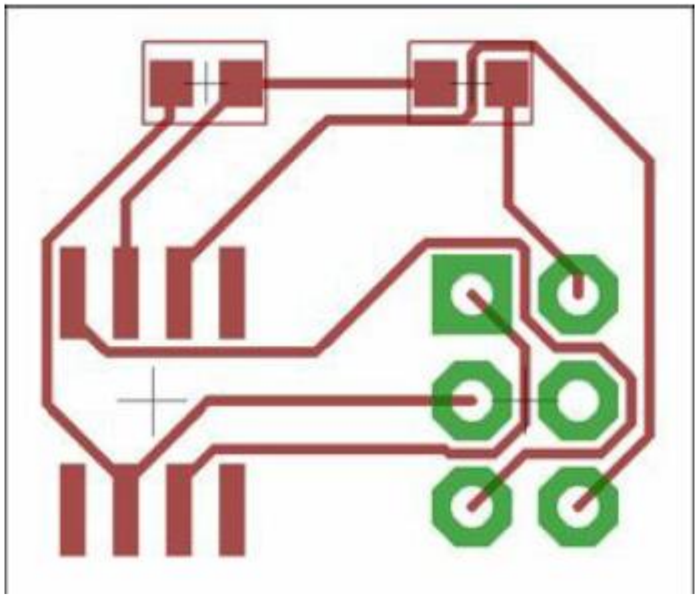


Figure 5: The Routed Inverting Amplifier Circuit